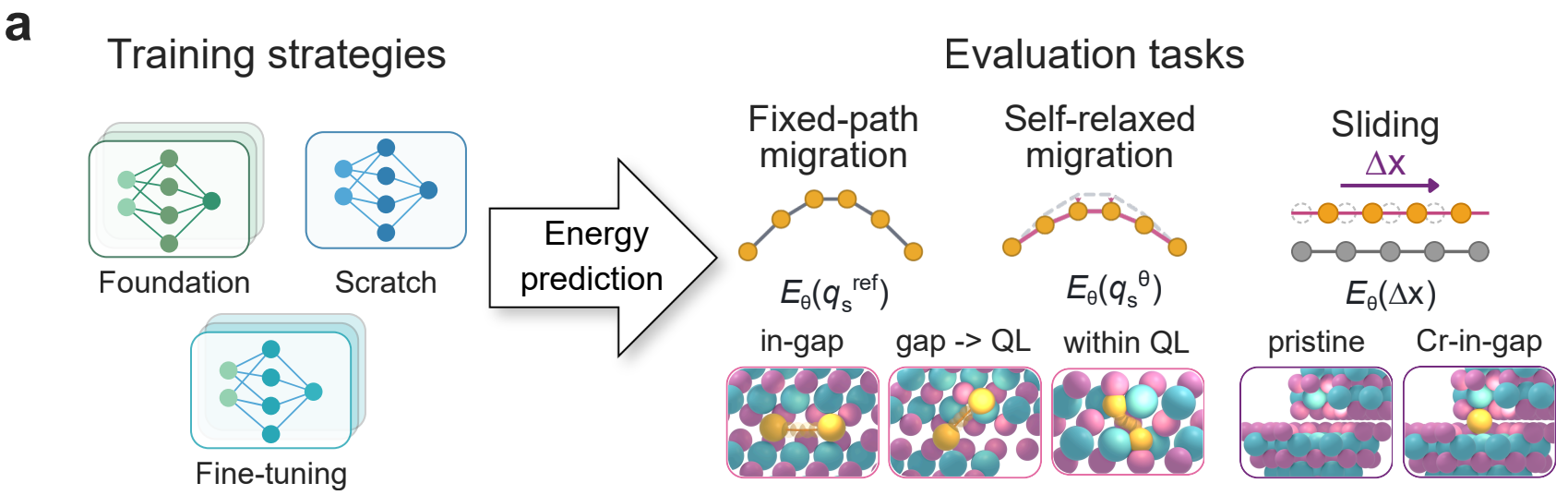
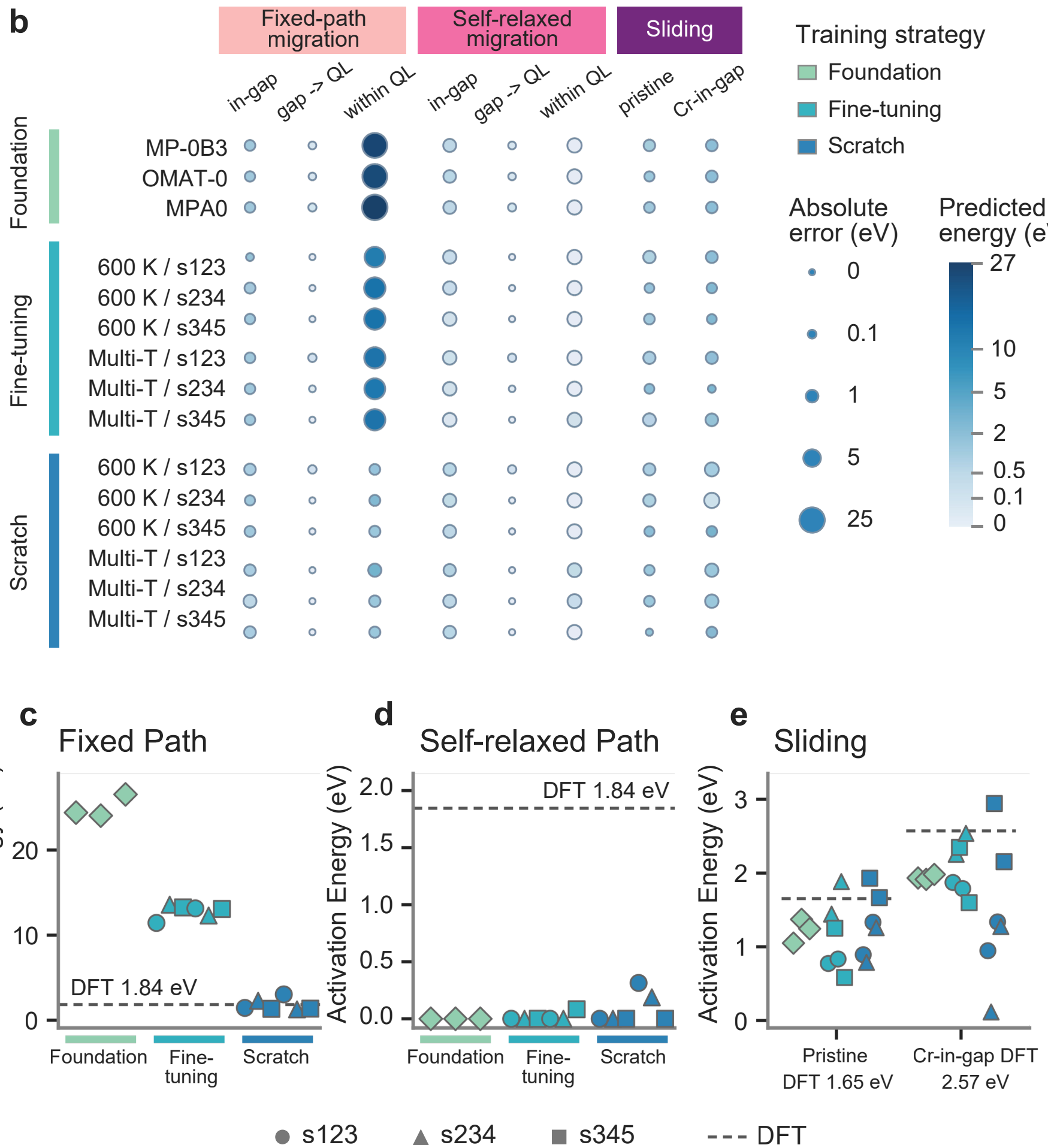

a
Training strategies
Foundation
Scratch
Fine-tuning
Energy prediction
Evaluation tasks
Fixed-path migration
Self-relaxed migration
Sliding
Δx
$E_\theta(q_s^{ref})$
$E_\theta(q_s^{\theta})$
$E_\theta(\Delta x)$
in-gap
gap -> QL
within QL
pristine
Cr-in-gap


b
Fixed-path migration
Self-relaxed migration
Sliding
in-gap
gap -> QL
within QL
in-gap
gap -> QL
within QL
pristine
Cr-in-gap
Foundation
MP-0B3
OMAT-0
MPA0
Fine-tuning
600 K / s123
600 K / s234
600 K / s345
Multi-T / s123
Multi-T / s234
Multi-T / s345
Scratch
600 K / s123
600 K / s234
600 K / s345
Multi-T / s123
Multi-T / s234
Multi-T / s345
Training strategy
Foundation
Fine-tuning
Scratch
Absolute error (eV)
0
0.1
1
5
25
Predicted energy (eV)
27
10
5
2
0.5
0.1
0
c
Fixed Path
Activation Energy (eV)
20
10
0
DFT 1.84 eV
Foundation
Fine-tuning
Scratch
d
Self-relaxed Path
Activation Energy (eV)
2.0
1.5
1.0
0.5
0.0
DFT 1.84 eV
Foundation
Fine-tuning
Scratch
e
Sliding
Activation Energy (eV)
3
2
1
0
Pristine
DFT 1.65 eV
Cr-in-gap DFT
2.57 eV
s123
s234
s345
DFT